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Level 1 Physics, Earth and Space Science

1.4, 2025

92047 Demonstrate understanding of a physical system using energy concepts

Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Demonstrate understanding of a physical system using energy concepts.	Explain a physical system using energy concepts.	Analyse a physical system using energy concepts.

You should attempt ALL the questions in this booklet.

In your answers use clear numerical working, words, and / or diagrams as required.

Numerical answers should be given with an appropriate SI unit.

If you need more space for any answer, use the extra space provided at the back of this booklet and clearly number the question.

Check that this booklet has pages 2 – 11 in the correct order and that none of these pages is blank.

YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE ASSESSMENT.

TOTAL

You may find the following formulae and data useful.

$$E_k = \frac{1}{2}mv^2 \quad \Delta E_p = mg\Delta h \quad g = 10 \text{ N kg}^{-1} \quad W = Fd$$

$$E(\text{thermal}) = mc\Delta T \quad E(\text{thermal}) = mL$$

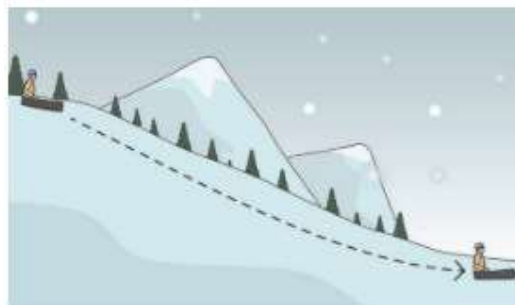
$$V = IR \quad P = IV \quad \Delta E = P\Delta t$$

$$c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$$

QUESTION ONE

Zac visits the mountains this winter to go sledding. He starts at the top of an icy incline and rides his sled down to the bottom. His sled reaches its top speed at the bottom of the incline.

Assume air friction and friction between the sled and the surface it is sliding on are negligible.



- (a) Describe the energy changes during his sled ride down the incline.

- (b) Zac and his sled have a combined mass of 70 kg. The top speed that Zac reaches at the bottom of the incline is 12 m s^{-1} .

Show that Zac has 5040 J of kinetic energy at the bottom of the incline, then calculate the vertical height of incline.

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- (d) Zac needs to carry the sled back up the hill to his car. He has two paths to choose from to get there. One path is a slow steady incline, while a second path has a set of steep stairs. Both have a vertical height of 3 m.

It takes Zac 30 s to walk up the incline, but only 10 s to climb the stairs.

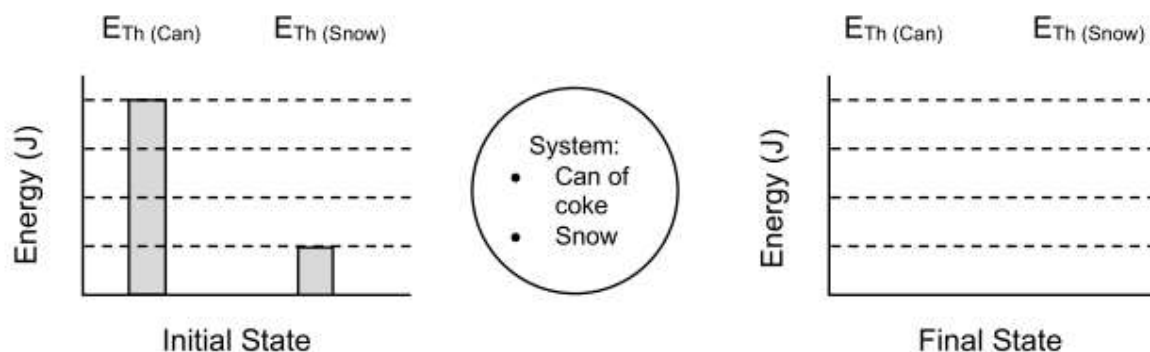
Zac and his sled have a combined mass of 70 kg.

Carry out calculations to determine which path will feel easier.

QUESTION TWO

Zac is staying at the top of the mountain at a ski resort. After a hard day of sledding, Zac returns to his room for a cold drink and a hot spa.

- (a) Zac takes a warm can of Coke and places it outside his room in the snow to cool it down. Complete the energy bar chart for the can and snow to show how thermal energy (E_{Th}) is transferred, and the can cools down.



There is an outdoor spa pool that is just outside Zac's room.

- (b) The spa pool holds 500 L of water (500 kg), and was originally filled with 20 °C water. A simple heater circuit heats up the water.

If the heater has a power output of 960 W, calculate how many hours it took to heat the spa pool water up to 36 °C.

The specific heat capacity of water is 4200 J kg⁻¹ °C⁻¹.

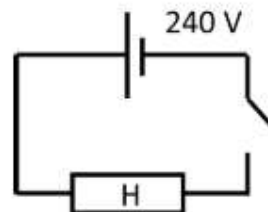
- (c) Describe the movement of the water particles in the hot spa pool (liquid water), and in the snow surrounding the spa pool (ice/solid water).
By comparing the forces between water particles in the liquid state to those in the solid state, explain why the particles move in this way in hot water and in snow/ice.

- (d) The heater is turned off, and the spa pool water slowly starts to cool down in the cold mountain air. Zac puts the spa pool lid on to minimise this cooling. The lid is made of a thick insulating material with a bottom surface cover made of a silvery material.
- Describe the heat transfer mechanisms that cause heat from the surface of the water to be transferred into the air above the lid.
 - Explain, using physics principals, how this lid minimises these heat transfer mechanisms, and retains the heat in the spa pool as long as possible.

QUESTION THREE

The next morning before Zac goes out to his final day of sledding, he gets up early and makes himself a hot drink using a 1200 W electric kettle.

The kettle circuit to heat up the water consists of a 240 V power supply, a switch, and a heating element.

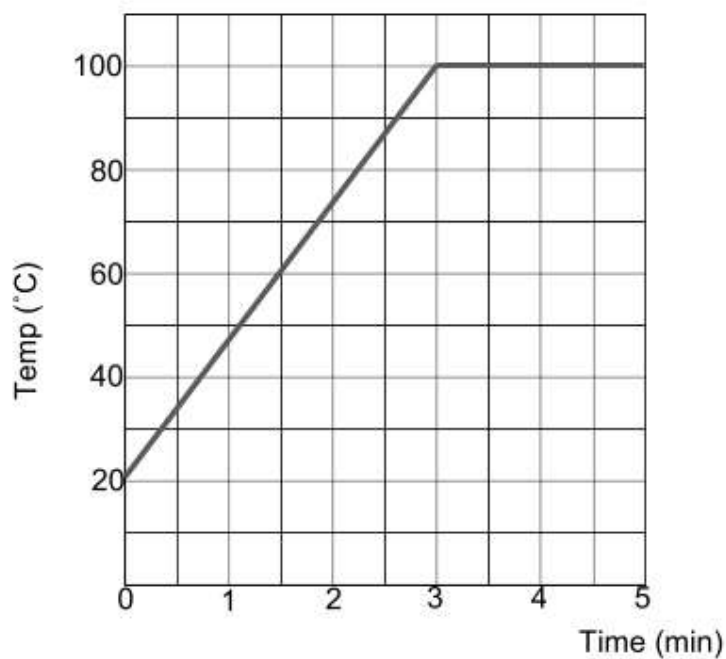


The specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

- (a) Zac puts 750 mL of water (750 g) into the kettle. The temperature of the water is 20°C . Calculate how much heat energy is required to raise the temperature of the water to 100°C .

- (b) By first calculating the current in the circuit, calculate the resistance of the heating element.

- (c) Zac switches the 1200 W kettle off after five minutes. The graph below shows how the water temperature in the kettle changed over time while it was switched on.



Explain why the temperature of the water in the kettle remains at 100 °C for the final two minutes of the graph, despite the kettle still being on.

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Extra paper if required.
Write the question number(s) if applicable.

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